MARK SCHEME · TEACHER COPY

10BR · Chapter 1: Data Representation

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 73

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · May/June 2024 Paper 12, original Q1 (a),(b),(c),(d) [6 marks]

Question	Answer	Marks
1(a) i)	(1) byte	1
1(b)	8192	1
1(c)	(1) Tebibyte // TiB	1
1(d)	**One** mark for each correct stage of working (**max** 2): • 512 × 512 • 262 144 * 2 // multiplied by 16 and divided by 8 • 524 288/1024 **One** mark for the correct answer: 512 (KiB)	3

Question 2 · May/June 2024 Paper 12, original Q3 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks
2(a) i)	Hexadecimal	1
2(b)(i)	1010 110010 11001001	3
2(b)(ii)	**Two** from: • Computers use **logic gates/switches** … • … that only process the values 1 and 0 // that only have two states	2
2(c)	**One** mark for evidence of working, for example 2 carries **One** mark for each correct nibble (Max 2) 1 1 0 0 1 1 0 0 0 0 0 1 1 0 0 1 1 0	3

2(d)	**One** marking for evidence of working For example, flip and add **One** mark for correct binary 11100000	**2**

Question 3 · May/June 2025 Paper 12, original Q1 (c),(d),(f),(g) [8 marks]

Question	Answer	Marks
3(a)	• 10011 • 11100110	**2**
3(b)	• 0011 0101 • 1000 1010 1101	**2**
3(c)	Any **two** from: • The **result** is greater than 255 // The **result** is too large. • Cannot be stored in **8 bits** // Cannot be stored in the number of bits available for the **register**	**2**
3(d)	**One** mark for a correct working method e.g. flip and add **One** mark for correct answer 11101010	**2**

Question 4 · May/June 2025 Paper 12, original Q2 (b),(c),(d) [8 marks]

Question	Answer	Marks
4(a)(i)	3	**1**
4(a)(ii)	2	**1**
4(b)(i)	32 bits are used to represent **each/a/one** colour in the image // 2^{32} (approx. 4.3 billion) different colours can be used/are available for the image	**1**
4(b)(ii)	Any **two** from: • The size of the image **file** increases. • … as the number of bits used to represent/store a colour has increased	**2**

4(c)	Any **three** from: • The size of the file is reduced without **permanently** removing any data. • A compression algorithm is used. • … such as Run length encoding/RLE. • Repeating **pixels** are grouped/identified … // **Patterns** are identified … • … and stored with the number of times they are repeated. • … and indexed	**3**

Question 5 · May/June 2026 Paper 11, original Q1 (a),(b),(c) [3 marks]

Question	Answer	Marks	Guidance
5	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.		
5(a)	A	**1**	If more than one box is ticked, no mark can be given.
5(b)	3	**1**	Mark the first answer given. Ignore any reference to unit. If no answer is written in the answer space an obvious correct answer can be awarded in the working space.
5(c)	C	**1**	If more than one box is ticked, no mark can be given.

Question 6 · May/June 2026 Paper 11, original Q3 (a),(b),(c),(d),(e),(f) [9 marks]

Question	Answer	Marks	Guidance

6	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.		
6(a)	**One** from: • Denary // decimal • Two's complement	**1**	Mark the first answer given. Accept other system outside syllabus e.g. octal, BCD BOD Base-10
6(b)	0010 1010	**1**	Mark the first answer given. Correct number of leading zeros must be present If no answer is written in the answer space an obvious correct answer can be awarded in the working space.
6(c)	0001 0000 0001	**1**	Mark the first answer given. Correct number of leading zeros must be present If no answer is written in the answer space an obvious correct answer can be awarded in the working space.
6(d)	71	**1**	Mark the first answer given. If no answer is written in the answer space an obvious correct answer can be awarded in the working space.

Total: 73 marks

6(e)	2D8	**1**	Mark the first answer given. If no answer is written in the answer space an obvious correct answer can be awarded in the working space.
6(f)	**One** mark for each correct nibble (MAX 2) **One** mark for the three correct carries as working **One** mark for one overflow bit For example: 1 1 1 1 0 0 1 1 1 0 1 + 1 1 1 1 0 0 0 0	**4**	Carries can be shown above/below. Award where seen as long as for the correct columns. BOD 4 carries shown if the overflow is shown as a carry first, then as overflow. Ignore the crossing out of carries. Award the number of carries seen.

Question 7 · May/June 2026 Paper 11, original Q8 (a),(b) [10 marks]

Question	Answer	Marks	Guidance
7(a)	Any **three** from: • A character set is used … • … that contains **all** the characters (that can be represented) • It can use ASCII/Unicode • **Each/a** character is given/has a (binary) value	**3**	MP4 – BOD if code used in place of value
7(b)(i)	One mark for each correct term or definition. **term** / **description** pixel – the smallest unit of an image that is made up of one colour sound sample – a (single) measure/representation of the amplitude/height of a sound wave (image) resolution – the dimensions of an image colour depth – the number of bits that are used for each/a colour/pixel (in an image) // the (maximum) number of colours that can be represented (in an image)	**5**	Colour depth – 'the number of colours in the image' is not accurate

	Sample/sampling rate/frequency	the number of samples that are recorded each second		
7(b)(ii)	Lossy // by example e.g. motion estimation Lossless // by example e.g. RLE		**2**	Mark the first answer given. Do not accept file formats e.g. MP4, ZIP BOD lossyless for lossless

Question 8 · May/June 2026 Paper 12, original Q1 (b),(c) [4 marks]

Question	Answer	Marks	Guidance
8	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.		
8(a)	Bit(s)	**1**	Mark first answer given.
8(b)(i)	A	**1**	If more than one box ticked, no mark can be awarded.
8(b)(ii)	00100000	**1**	Mark first answer given. Leading zeros must be present. If the answer line is left blank, then an obvious correct answer can be awarded in the working space.
8(b)(iii)	11110010	**1**	Mark first answer given. If the answer line is left blank, then an obvious correct answer can be awarded in the working space.

Question 9 · May/June 2026 Paper 12, original Q3 (a),(b),(c),(d),(g) [14 marks]

Question	Answer	Marks	Guidance

9	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.		
9(a)	D	**1**	If more than one box ticked, no mark can be awarded.
9(b)	• 0001 0101 0010 • 0011 0000 1111	**2**	Only penalise missing leading zeros once. If the answer line is left blank, then an obvious correct answer can be awarded in the working space.
9(c)	• 475 • AE8	**2**	Mark first answer given. If the answer line is left blank, then an obvious correct answer can be awarded in the working space.
9(d)	• 162 • 260	**2**	Mark first answer given. If the answer line is left blank, then an obvious correct answer can be awarded in the working space.
9(e)(i)	**Two** from: • They want to retain the **original** version (of the video) // **Original** version (of the video) can be recovered • (The videos) will be the highest/high **quality** // The **quality** will not be reduced/will be retained // Artifacting will be less likely • Storage space is not a concern // File sizes are small enough (with lossless)	**2**	Mark whole answer. Accept MPs in reverse e.g. lossy would reduce the quality of the video MP1 – It is NE to just say no data will be permanently removed/data is restored. This is NE for the question, it is just saying what lossless does. MP1 – 'closer/similar to the original' is inaccurate, it is not closer to it, it will be the same as original when decompressed.

9(e)(ii)	**One** from: • Smaller **file** size • The videos would use less storage/memory space • The time taken to transmit the videos would be shorter // Faster transmission/upload/download • Less bandwidth would be used to transmit the videos • Video would be less likely to buffer when streaming	**1**	MP3 – BOD faster loading time
9(e)(iii)	**Four** from: • A compression algorithm is used • … such as RLE • Repeating **frames/pixels** are identified … • Repeating **samples** are identified (for sound) … • … and indexed // by example • Eliminating redundancy // Uses temporal redundancy • … by recording the differences between each frame/amplitude (for sound)	**4**	MP3 and MP4 - Repeating data/patterns is TV MP5 is dependent on either MP3 or MP4. Accept others that are correct for other lossless methods, for example Huffman encoding.